

Exercise 6 - Ab initio and semi-empirical molecular dynamics of a water cluster

Both runs use the ORCA %md module on a cluster cut from an ice crystal, at 400 K with a Nosé–Hoover chain thermostat (time constant 10 fs) and a time step of 0.5 fs, the largest step the manual recommends for systems with hydrogen atoms. The 50 Å sphere was taken as the diameter, so Cell Sphere was set to a radius of 25 Å, centred on the cluster because the input coordinates are not centred on the origin. The sphere is a soft repulsive wall that stops evaporated molecules from flying away. The clusters themselves have radii of only 6.4 Å (a) and 13.7 Å (b), so the wall does not affect the cluster itself and a 50 Å radius would give the same result. Keywords: ! MD PBE def2-SVP def2/J for (a) and ! MD XTB2 for (b).

Setup

Table 1. The two simulations.

	(a) PBE/def2-SVP	(b) GFN2-xTB
Water molecules	20	128
Atoms	60	384
Steps × 0.5 fs	200 = 100 fs	5000 = 2.5 ps
Wall time (1 core)	4.8 h	5.6 h
Time per step	86 s	4.1 s

Trajectory in Avogadro

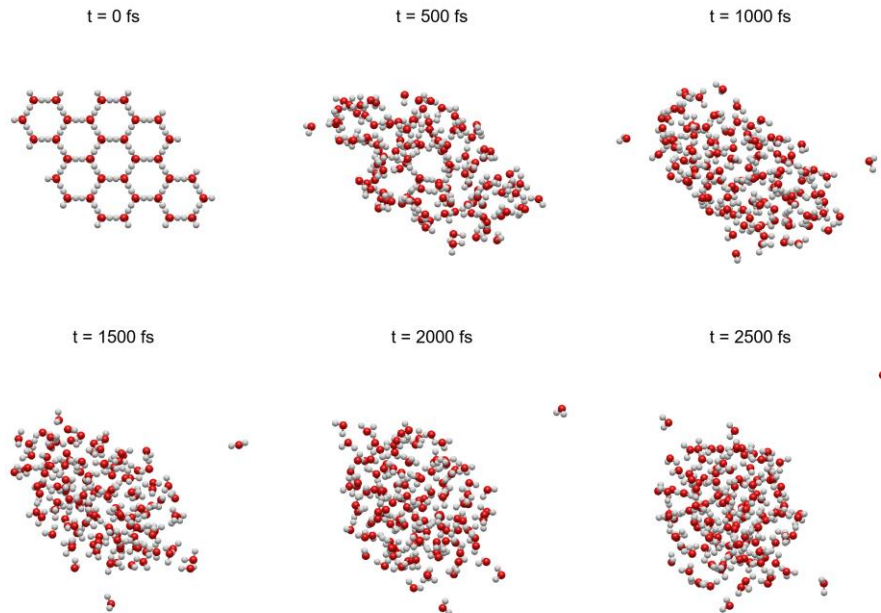
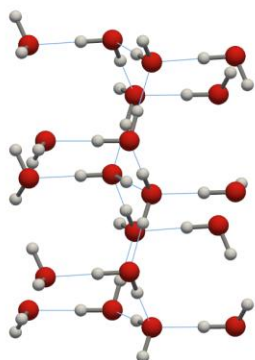


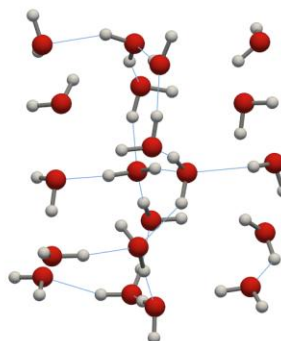
Figure 1. The xTB trajectory (b) opened in Avogadro 2, frames at 500 fs intervals. Red: O, white: H. Because the movie export of Avogadro does not work, the trajectory was displayed frame by frame in Avogadro (every 100th MD step, 51 frames) and the rendered frames were joined into an animated GIF, which is sent with this report as [HW2_ZhangXubo_Ex6_water_MD_XTB_Avogadro.gif](#).

Structures at the beginning and at the end

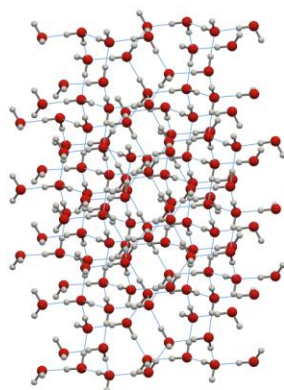
(a) PBE/def2-SVP, 20 H₂O
start: $t = 0$ fs



(a) PBE/def2-SVP, 20 H₂O
end: $t = 100$ fs



(b) GFN2-xTB, 128 H₂O
start: $t = 0$ fs



(b) GFN2-xTB, 128 H₂O
end: $t = 2500$ fs

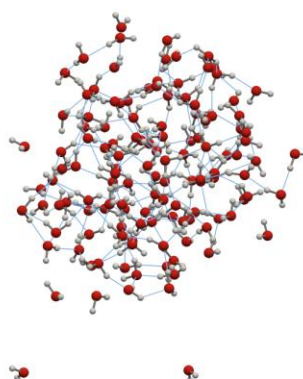


Figure 2. First and last frame of each trajectory. Red: O, white: H, grey sticks: covalent O–H bonds, thin blue lines: hydrogen bonds ($H\cdots O < 2.2$ Å). Same viewing direction and scale within each row.

Temperature and structure along the trajectories

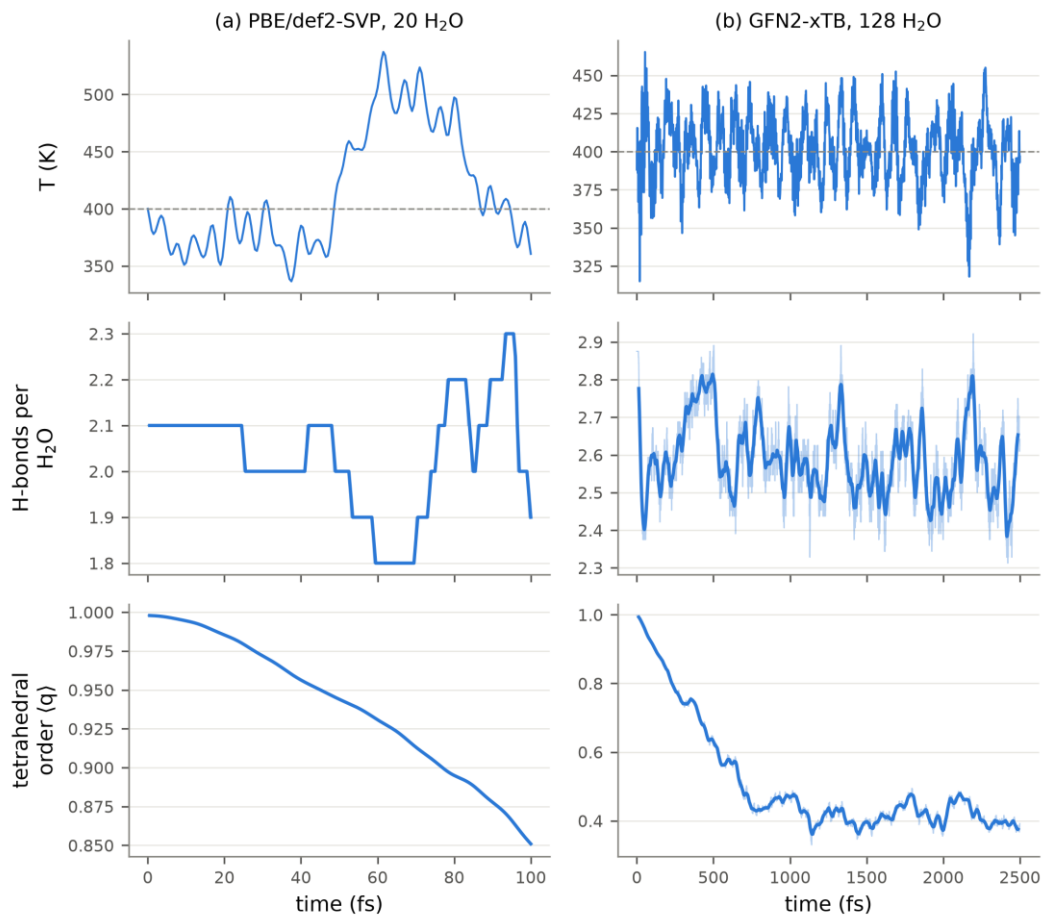


Figure 3. Top: instantaneous temperature (dashed line: the 400 K set point). Middle: hydrogen bonds per water molecule ($O\cdots O < 3.5 \text{ \AA}$ and $\angle(H-O\cdots O) < 30^\circ$). Bottom: tetrahedral order parameter q averaged over the molecules that start with four neighbours within $3.2 \text{ \AA}$ (2 of 20 and 48 of 128). Thin lines are per-step values, thick lines a running average.

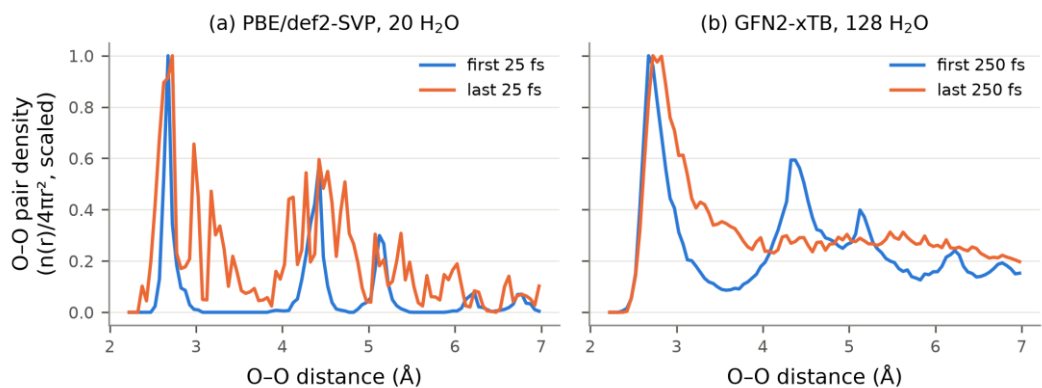


Figure 4. O-O pair distance distribution, $n(r)/4\pi r^2$, at the beginning and at the end of each run, scaled to the same maximum.

What the simulations show

In the xTB run the ice cluster melts, and this shows up in the structure rather than in the temperature. The tetrahedral order of the interior molecules drops from 0.90 to 0.39 within the first ~800 fs and then stays there. In the O–O distribution the second-neighbour peak at 4.4 Å disappears and the minimum at 3.5 Å fills in (Figure 4b). The number of hydrogen bonds hardly changes (2.56 → 2.53 per molecule), so the molecules keep their hydrogen bonds but lose the tetrahedral arrangement and the long-range order of the crystal.

The 100 fs DFT run is too short to see this: q only falls from 0.99 to 0.88 and the pair distribution still has the ice peaks. Hydrogen bonds in water rearrange on a picosecond time scale, so 100 fs only samples vibrations around the starting structure.

In the xTB run some molecules evaporate. The largest O distance from the cluster centre grows from 13.0 Å to 23.6 Å, close to the 25 Å wall, so without the confining sphere these molecules would leave the cluster.

The thermostat keeps the average temperature at the set point: over the second half of the xTB run the mean is 399 K with a standard deviation of 22 K. The 60-atom DFT cluster fluctuates by about ± 50 K, because relative temperature fluctuations scale as $1/\sqrt{N}$ and are large in small systems.

What else can be obtained from the trajectories

Besides the melting, the trajectory and the per-step energies that ORCA writes can be used for:

Structure: radial distribution functions $g(r)$ for O–O, O–H and H–H and the coordination numbers from their integrals, hydrogen-bond distance and angle distributions, and the size and shape of the cluster.

Dynamics: the mean square displacement and the diffusion coefficient, hydrogen-bond lifetimes and reorientation times, and the velocity autocorrelation function, whose Fourier transform gives the vibrational density of states. The dipole autocorrelation function gives the IR spectrum including anharmonic and temperature effects, which a harmonic frequency calculation as in Exercise 1 cannot provide.

Thermodynamics and checks: kinetic and potential energy distributions (the heat capacity follows from their fluctuations), the evaporation rate, and the drift of the conserved energy as a check of the integration, here $-3.2 \text{ kcal mol}^{-1}$ over 2.5 ps for xTB and $+0.03 \text{ kcal mol}^{-1}$ over 100 fs for DFT.

Snapshots from the trajectory can also serve as starting structures for further calculations, for example spectra of solvated molecules averaged over the sampled configurations.